

Task 1. Install Ubuntu Server on a VirtualBox VM or your physical machine, then verify the successful installation.

Step 1: Create the Virtual Machine

Open VirtualBox and create a new virtual machine. Select Linux/Ubuntu as the operating system and assign suitable RAM, CPU, and virtual disk resources.

Step 2: Attach Ubuntu Server ISO

Attach the Ubuntu Server ISO file to the virtual machine's optical drive and start the VM.

Step 3: Install Ubuntu Server

Follow the installer, configure the keyboard, network, storage, hostname, username, and password, and complete the installation.

Step 4: Restart and Login

Restart the machine after installation. Log in with the created user account and confirm that the Linux terminal is available.

Result

Ubuntu Server was installed successfully and the system reached the Linux command-line environment. The server is ready for basic administration and file-system practice.

Verification

```
whoami
hostname
uname -a
```

Task 2. Create a directory named MyMusic and create three empty files inside it: playlist.txt, songs.txt, and artists.txt.

Step 1: Create the directory

Use the mkdir command to create a new directory named MyMusic.

```
mkdir MyMusic
```

Step 2: Enter the directory

Move into the newly created directory.

```
cd MyMusic
```

Step 3: Create the empty files

Use touch to create the three required empty text files.

```
touch playlist.txt songs.txt artists.txt
```

Step 4: Verify the files

List the directory contents.

```
ls -l
```

Result

The MyMusic directory contains the three required files: playlist.txt, songs.txt, and artists.txt. The files are empty because they were created with touch.

Task 3. Navigate to the MyMusic directory, list all files inside it, and display the current working directory path.

Step 1: Navigate to MyMusic

From the home directory, enter the MyMusic directory.

```
cd ~/MyMusic
```

Step 2: List the files

Use ls to display the contents of the current directory.

```
ls
```

Step 3: Display the current path

Use pwd to display the absolute path of the current working directory.

```
pwd
```

Expected Output

```
artists.txt playlist.txt songs.txt  
/home/<username>/MyMusic
```

Result

The ls command displays all three files and pwd confirms that the current working directory is the MyMusic directory under the user's home directory.

Task 4. Compare the file-system structure of Ubuntu Linux with a Windows PC and write three key differences.

The following comparison identifies three important differences between the Linux and Windows file-system structures.

Difference	Ubuntu Linux	Windows
Root / drive structure	Uses one unified root directory: /. All directories are organized below the root.	Uses drive letters such as C:, D:, and E: to identify storage volumes.
User files	Normally stored in /home/<username>/.	Normally stored in C:\Users\<username>\.
System configuration	Many system-wide configuration files are stored in /etc/.	Windows system files are mainly located under C:\Windows, with configuration managed through Windows-specific system locations.

Result

Linux uses a single hierarchical directory tree beginning at /, while Windows commonly separates storage into drive letters. Linux also uses locations such as /home and /etc, whereas Windows commonly uses C:\Users and C:\Windows.

Task 5. Use the man command in Linux to find out what the ls command does, and write one new ls option learned from the manual.

Step 1: Open the ls manual

Use the man command to display the manual page for ls.

```
man ls
```

Step 2: Understand the command

The ls command is used to list information about files and directories.

Step 3: Option learned: -l

The -l option displays a long listing containing additional information such as permissions, ownership, file size, and modification time.

```
ls -l
```

Step 4: Another useful option: -a

The -a option includes hidden files and directories in the listing.

```
ls -a
```

Answer

The ls command lists files and directories. From the manual, the -l option was learned. It displays a detailed long-format listing of directory contents. The -a option is also useful because it shows hidden files and directories.

Result

The man command was successfully used to study the ls command and its available options.

Task 6. Conclusion

This practical introduced basic Linux server administration through Ubuntu Server. The required MyMusic directory and three empty files were created using Linux commands, directory navigation and path verification were practiced, and the Linux and Windows file-system structures were compared. The man command was also used to understand the ls command and its options. These commands provide a foundation for further Linux server management and cybersecurity practical work.

Commands Used

```
mkdir MyMusic  
cd MyMusic  
touch playlist.txt songs.txt artists.txt  
ls -l  
cd ~/MyMusic  
ls  
pwd  
man ls  
ls -a
```